Exec. graph	Cases generated	**Stage 2:** program	**Stage 2:** annotation	**Stage 3:** gem5 harness
Store ↓*rf* Load *store, aliasing load*	Predicted no alias, actually aliases (false negative). Own mode; all branches resolved.	S and L share an 8-byte slot; with an address register, S is re-addressed to L's runtime address. Inline asm stops the compiler forwarding S to L.	(S, L) PC pair per aliasing load	STL-hooked build: S's address generation held 5000 cycles; L's dependence prediction forced to no-alias. *Window check not yet built.*
Branch *unresolved branch*	Mispredicted not-taken; mispredicted taken. *Correctly predicted: not generated.*	Jump on a constant register (`movq $1` / `xorq`). The architectural path goes to the end block, so the next instruction runs only speculatively.	Mode; predicted target = next PC	Force the direction reaching the next instruction; hold resolution s_U=5000 cycles. Gate: unresolved at transmission; invalid unless mispredicted and held.
Branch *r* *resolved branch*	Correctly predicted not-taken. *Taken: not generated.*	Jump on a zeroed register (`xorq`): falls through, never takes its target.	Mode	Force the architectural direction; resolution delay swept over {0, 500, 2500} cycles per branch, all combinations. Invalid if it mispredicts.
lc *c* *fnc* *commit boundary*	lc: last committed; fnc: first that must stay uncommitted (Alloy instructions only).	Global labels on lc and fnc.	lc and fnc PCs	Window: lc retired < transmission < fnc retires (measured, not forced). Commit stall on fnc: 0 cycles.
t *target transmitter*	One per test: load, store, branch, multiply or divide.	Branch: `testq+jz` on the transmitted value, architecturally taken (`nop` after it if last). Divide: a zero divisor faults (other divides get `orq $1`).	PC; kind selects the checker	Branch: forced to mispredict, no stall. A divide fault at the transmitter counts as a transmission.
every instruction	All.	Inline-asm block per instruction with a global PC label; memory and live-in registers zeroed, then `mfence`; address operands index a probe array; clang `-O2`.	Labels resolved to x86 addresses	No hook: measured from the O3PipeView, LSQUnit and Commit traces.

Glyphs use the execution-graph syntax of Fig. (a): boxes are instructions in program order, top to bottom (empty box: any instruction); rf = reads-from; flags r resolved, c committed, t target transmitter. Each glyph shows only the flags its row is about: a Branch or Store without r is unresolved.